

Blue Sky Modeling

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Research questions

RQ1

Within monitored Bluesky and custom-feed top-K panels, after accounting for timing and panel context, do semantically equivalent or near-duplicate posts receive systematically unequal observed exposure opportunities? If so, which observable post-, author-, feed-, and viewer-level factors are associated with the remaining divergence?

RQ2

Within defined political topics and event clusters on Bluesky, do monitored top-K feed panels allocate observed, rank-weighted exposure across frames in proportion to observed frame supply? If not, how much frame disparity remains after timing and declared-objective adjustments, and how much is more consistent with dynamic or contestable amplification?

Turn ranking position into exposure opportunity

$$w_j = \frac{1}{\log(1 + j)}$$

Front positions count more than later positions.

$$E_p = \sum_{m=1}^{n_p} w(r_m).$$

n_p is the number of monitored top-K appearances of post p , and r_m is the rank of its m th appearance. This is the main exposure definition

RQ1: same content, different exposure ?

Take one duplicate or near-duplicate cluster and call it C . Its total observed exposure is:

$$E(C) = \sum_{p \in C} E_p.$$

The observed share of post p inside C is:

$$P_p = \frac{E_p}{E(C)}.$$

Example

Suppose one duplicate cluster contains three posts A, B, C:

- A appears three times, at ranks 1, 2, 6.
- B appears twice, at ranks 2, 3.
- C appears once, at rank 5.

Using $w(r) = 1 / \log_2 (1 + r)$, we have $w(1) = 1$, $w(2) = 0.6309$, $w(3) = 0.5000$, $w(5) = 0.3869$, $w(6) = 0.3562$.

$$EA = 1 + 0.6309 + 0.3562 = 1.9871$$

$$EB = 0.6309 + 0.5000 = 1.1309$$

$$EC = 0.3869$$

Example

The cluster total is $E(C) = EA + EB + EC = 1.9871 + 1.1309 + 0.3869 = 3.5049$

$$P_p = \frac{E_p}{E(C)}.$$

$$(P_A, P_B, P_C) = \left(\frac{1.9871}{3.5049}, \frac{1.1309}{3.5049}, \frac{0.3869}{3.5049} \right) = (0.5670, 0.3227, 0.1104)$$

Why different exposure?

Some factors are ok.

Some are not....

What are the factors that drives the different exposures among near duplicate posts?

- Is it post created time?
- Is the author profile?
- Is the the early trajectory of a post?
- ... and more

There's way too many factors

Do one decomposition at a time..

$$P_p = \frac{E_p}{E(C)}.$$

The model-implied share is

$$\hat{P}_p = \left[\text{softmax}((\eta_q)_{q \in C}) \right]_p$$

The residual gap for one duplicate cluster is

$$D(C) = \frac{1}{2} \sum_{p \in C} |P_p - \hat{P}_p|.$$

one decomposition at a time..

Ideally, same post A and B and C should have the same exposure

$$\hat{P}_p^{(0)} = \frac{1}{|C|} \quad \hat{P}^{(0)} = (1/3, 1/3, 1/3).$$

$$(P_A, P_B, P_C) = \left(\frac{1.9871}{3.5049}, \frac{1.1309}{3.5049}, \frac{0.3869}{3.5049} \right) = (0.5670, 0.3227, 0.1104)$$

$$D_0 = \frac{1}{2} \left(|0.5670 - 0.3333| + |0.3227 - 0.3333| + |0.1104 - 0.3333| \right) = 0.2336.$$

Maybe time is the factor that cause difference?

Suppose the comparison time is such that the post ages are 10, 6, and 2 hours, and use the simple timing score

$$f(\Delta t) = 0.15 \Delta t.$$

$$\eta^{(1)} = (1.5, 0.9, 0.3), \quad \hat{P}^{(1)} = \text{softmax}(\eta^{(1)}) = (0.5405, 0.2967, 0.1628).$$

The New Gap:

$$D_1 = \frac{1}{2} \left(|0.5670 - 0.5405| + |0.3227 - 0.2967| + |0.1104 - 0.1628| \right) = 0.0524.$$

Maybe some posts fits better in that particular feed?

Let surface fit give a bonus only to post B, for example because B better matches the surface's stated objective. Set the surface bonuses to $(0, 0.1, 0)$. Then:

$$\eta^{(2)} = (1.5, 1.0, 0.3), \quad \hat{P}^{(2)} = (0.5242, 0.3179, |0.1579),$$

$$D_2 = \frac{1}{2} \left(|0.5670 - 0.5242| + |0.3227 - 0.3179| + |0.1104 - 0.1579| \right) = 0.0475.$$

Maybe author profile is a factor?

Now let both A and B have stronger author-profile signals than C, for example because they come from more established accounts. Use the simple author bonuses $(0.1, 0.1, 0)$. Then:

$$\eta^{(3)} = (1.6, 1.1, 0.3), \quad \hat{P}^{(3)} = (0.5322, 0.3228, 0.1450),$$

$$D_3 = \frac{1}{2} \left(|0.5670 - 0.5322| + |0.3227 - 0.3228| + |0.1104 - 0.1450| \right) = 0.0348.$$

Maybe post early trajectory

...

RQ2: same topic, different frames

Now move from duplicate posts to a political topic or event cluster and call it H . Let F denote one frame inside that topic cluster:

$$\text{Frames}(H) := \{F \subseteq H : F \text{ is one observed frame inside } H\}.$$

The total exposure of frame F is

$$E_F = \sum_{p \in F} E_p.$$

Example: Epstein

Topic of Interest: Epstein-related content.

Framing Perspectives:

- Democrats aim to highlight evidence implicating Republican leaders (e.g., Trump).
- Republicans aim to highlight evidence implicating Democratic elites.

Objective:

- Quantify the share of total exposure received by each framing perspective within the Epstein topic.

Exposure of a frame

The total exposure of frame F is:

$$E_F = \sum_{p \in F} E_p.$$

The observed exposure share of frame F is:

$$P_F = \frac{E_F}{\sum_{G \in H} E_G}.$$

Supply of a frame

How many post of frame F are there:

$$N_F^{\text{sup}} = \sum_{p \in U_H} \mathbf{1}[z_p = F].$$

The observed supply share of frame F is:

$$S_F = \frac{N_F^{\text{sup}}}{\sum_{G \in H} N_G^{\text{sup}}}.$$

The frame disparity of exposure within a topic H

$$D(H) = \frac{1}{2} \sum_{F \in H} |P_F - S_F|.$$

Hypothesis...

H1(1): Even if timing is controlled, the duplicate conditional exposure gap is still significantly greater than zero

H1(2): Early trajectory and(or) author profile has a effects on a post's future exposure

H2(1): in the same event cluster, there is a systematic offset between frame exposure share and frame supply share.

H2(2) the intensity of frame level discrimination is different under different feed types, viewer modes or moderation environments.